Health Technology Ecosystem

RFI Response
CMS-0042-NC

June 13, 2025

Submitted To:
Centers For Medicare And Medicaid Services
Office Of Acquisition & Grants Management
7500 Security Blvd
Baltimore, MD 21244-1850

Submitted By:
Bellese Technologies, Llc
10451 Mill Run Circle | Suite 400
Owings Mills, MD 21117
Uei: HXBBEPUC48F5
Cage Code: 5CLX2

This proposal includes data that shall not be disclosed outside of the government and shall not be duplicated, used, or disclosed—in whole or in part—for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of or in connection with the submission of this data, the government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the government's right to use information contained in this proposal if it is obtained from another source without restriction. The data subject to this restriction is contained on all sheets herein so designated.

Submission Cover Letter

June 13, 2025

Centers for Medicare and Medicaid Services
Department of Health and Human Services
7500 Security Blvd, Baltimore, MD 21244-1850

SUBJECT: CMS-0042-NC, Request for Information (RFI) - Health Technology Ecosystem

Bellese Technologies, LLC (Bellese) hereby submits our response to the solicitation for the Health Technology Ecosystem RFI for public comment. We appreciate the opportunity to contribute to the improvement of patient-centered healthcare in the US.

Bellese is a small digital services business with more than 13 years of experience supporting CMS through product, research, and engineering leadership, specializing in managing and delivering reliable, scalable, and secure digital products. With a focus on civic health tech, we have a deep understanding of how technology, government, medicine, public health, and the healthcare system converge. This expertise enables us to identify opportunities that enhance healthcare accessibility, navigability, and transparency for all Americans.

Having been embedded in CMS's digital ecosystem for over a decade, we have hands-on expertise in solving for the range of challenges CMS has with interoperability:

- As the current contract holder for the Quality Management and Review System (QMARS) development we are actively modernizing the system that facilitates resolution of Medicare patient complaints about provider care.
- Our involvement on Medicare Payment Systems Modernization (MPSM) included moving several complex service pricers from antiquated, brittle mainframe solutions to a modern, reusable cloud-based solution.
- We are currently developing and maintaining the Hospital Quality Reporting (HQR) system for Center for Clinical Standards and Quality (CCSQ), supporting over a dozen clinical quality reporting programs and computing Electronic Clinical Quality Measures (eCQMs) for the Inpatient Quality Reporting (IQR) and Outpatient Quality Reporting (OQR) programs. We continue to help CMS explore and plan the transition to digital quality measurement.
- Since 2019, we have attended all HL7 FHIR Connectathons and nearly all Working Group Meetings to represent CMS's interests. We have directly contributed to the relevant FHIR Implementation Guides through collaboration with authors based on our experience with HQR and we continue to participate in Connectathon testing activities.

As a result of our CMS engagements, we have a unique perspective on the needs, benefits, and challenges for both CMS and the healthcare providers. We believe that with aggressive and targeted actions CMS can enable a transformation of American health care where quality patient care is directed by well-informed patients. Our suggestions for these targeted actions are elaborated below in response to the RFI questions.

Respectfully,

Mark Krempley,
Chief Operating Officer

Table of Contents

Response to Questions from Section E. Technology Vendors, Data Providers, and Networks 4

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost? 4

TD-6. What unique interoperability functions does TEFCA perform? 4

TD-6a. What existing alternatives should be considered? 5

TD-11a. Should this capability be revised to specify standardized API requirements for EHI export? 5

TD-11c. Should CMS consider policy changes to support this capability's use? 5

TD-15. Regarding bulk FHIR APIs: 6

TD-15a. How would increased use of bulk FHIR improve use cases and data flow? 6

TD-15b. What are the potential disadvantages of their use? 6

Response to Questions from Section E. Technology Vendors, Data Providers, and Networks

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

Right now, the lack of a nationwide entity (providers, payers, vendors, etc.) directory causes distinct, complex processes and long onboarding times to access each individual provider system; makes accessing existing, federally-mandated healthcare data inefficient and inconsistent; and is a barrier to innovation due to lack of standardization. A nationwide, publicly accessible directory of FHIR endpoints would significantly enhance interoperability by simplifying access to health records, improving care continuity, and reducing redundant procedures by having all the information standardized in one easy-to-access format. The directory would also drastically improve interoperability in conjunction with the industry-wide implementation of the FAST Scalable Security for Registration, Authentication, and Authorization (SSRAA)–coming in 2026–to enable dynamic registration and access to data exchange APIs, reducing the need to spend time and money on bespoke integrations with every entity.

We believe CMS and ASTP/ONC should oversee the implementation and publication of a nationwide, publicly accessible directory of FHIR endpoints and entity organizational metadata, to advance their mission of fostering healthcare transformation to deliver care that can lower costs and improve care. By moving away from legacy, high-cost data exchanges will improve CMS' bottom line spending, reducing monthly expenditures for Medicare, Medicaid or CHIP beneficiaries. CMS already has access to a qualified pool of vendors capable of completing this project through the Agile Collaboration and Modernization Endeavors (ACME) Blanket Purchase Agreement (BPA).

We believe that through a modest investment by CMS, with nominal on-going maintenance burden, CMS has the opportunity to enable broad and impactful innovation in the private sector by unlocking the value that has already been created, in the private sector by prior legislation, but remains largely unrealized due to the undiscoverable nature of existing data and resources.

TD-6. What unique interoperability functions does TEFCA perform?

By designating Qualified Health Information Networks (QHINs), TEFCA creates a "network of networks," facilitating seamless data exchange among participants. This structure simplifies connectivity, reduces the need for multiple integrations, and promotes trust among entities by ensuring adherence to common standards and policies. By providing a unified approach to data exchange, TEFCA enhances the ability of patients, providers, and payers to access and share health information efficiently, improving care coordination, reducing administrative burdens, and supporting health initiatives.

TEFCA offers unique connectivity, such as its Record Locator Services which enables patients, their delegates, and healthcare providers to discover then acquire the patients healthcare records across diverse and disparate connected systems. We believe that CMS should leverage TEFCA

across its many data exchange use cases in order to provide a seamless data exchange experience for all participants and subparticipants, leading the industry in this new, innovative future.

TEFCA also supports FHIR-based exchange, with a roadmap to support point-to-point FHIR exchange, aligning it with CMS' goal to move to FHIR-based data exchange.

TD-6a. What existing alternatives should be considered?

While there are some potential alternatives, they are yet unproven and don't promote seamless interoperability on a national scale, where TEFCA can make that claim. CMS' support of TEFCA will demonstrate strength and leadership across industry, moving towards driving large-scale adoption of health management and care navigation applications, and reducing barriers to data access and exchange, promoting better health outcomes, and accelerating progress towards a patient-centric learning health system.

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

TD-11a. Should this capability be revised to specify standardized API requirements for EHI export?

Yes, the capabilities should be revised to specify standardized API requirements for EHI export. Lack of standardization is often a hindrance to the development of and innovation in Health IT; without a specific FHIR Implementation Guide, like the Argonaut Project's EHI Export API IG, varied implementations require bespoke, costly integrations. Given the wide variety of use cases enabled by an effective and universally available EHI export capability, CMS and ASTP/ONC should definitely refine the legislative requirements to specify a FHIR Implementation Guide.

TD-11c. Should CMS consider policy changes to support this capability's use?

Yes, CMS should implement policy changes to support standardized API requirements for EHI export. We believe that policy that focuses on providing guidance and technical requirements increases the ease of implementation within the Health IT community. As with previous legislation around interoperability, standards such as the United States Core Data for Interoperability (USCDI) and the recommended FHIR Implementation Guides, ensured that organizations could conform to the final rule(s) and build dependable testing for that conformance based on the IGs.

TD-15. Regarding bulk FHIR APIs:
TD-15a. How would increased use of bulk FHIR improve use cases and data flow?

Like the national directory, bulk FHIR APIs are a critical piece of interoperable functionality. Implementing Bulk FHIR APIs can significantly enhance data exchange by enabling efficient, large-scale data transfers. Bulk FHIR APIs allow for comprehensive analytics, leading to improved patient outcomes and streamlined healthcare operations. The asynchronous nature of bulk FHIR APIs reduces system load and latency, making it feasible to process extensive data requests without compromising performance. The asynchronous transfer of large data sets is an ideal solution for the transfer of patient data to quality organizations such as CMS, NCQA and others. The value of Bulk Data in this use case has been demonstrated by the NCQA Bulk FHIR Quality Coalition. This capability supports various bulk use cases, including population health management, clinical research, digital quality measurements, value-based care, and public health reporting.

TD-15b. What are the potential disadvantages of their use?

There are some potential disadvantages of bulk FHIR, such as:

- **Infrastructure and Scalability Concerns:** Implementing bulk FHIR APIs requires intensive infrastructure requirements capable of handling large volumes of data efficiently. Healthcare organizations may lack the necessary resources or technical expertise to support such infrastructure, leading to potential performance issues and system bottlenecks.
- **Data Privacy and Security:** Transferring large datasets increases the risk of data breaches and unauthorized access. Ensuring data privacy and security during bulk transfers is paramount, necessitating stringent security protocols and compliance with regulations).
- **Operational Challenges:** The asynchronous nature of bulk FHIR operations can result in inefficient data synchronization and consistency across systems. Additionally, varying levels of FHIR implementation among EHR vendors can lead to interoperability issues, hindering seamless data exchange.
- **Regulatory and Compliance Issues:** Lack of standardized guidelines for bulk FHIR API usage can result in inconsistent implementations and compliance challenges. Establishing clear regulatory frameworks and performance benchmarks is essential to promote uniform adoption and operational efficiency.

While there are some disadvantages, we believe that they are easily overcome by CMS and ONC establishing a regulatory framework for performance metrics.